

Max Marks: 70

Time Allowed: 2 hours

Name: _____

ERP: _____

Instructions:

- Answer the questions in the spaces provided on the question sheets.
- Extra sheets may be used for rough work, but make sure to write your name and ERP on them. Answers on extra sheets will not be graded.
- Understanding the questions is part of the exam, please do not ask for clarifications during the exam.

Question:	Expressions	Tracing	Debugging	Coding	Total
Marks:	12	28	10	20	70
Score:					

Question 1: Expressions 12 marks

(a) [12 marks]

For each of the expressions below, provide its C++ type (1/2mark) and value (1 mark). The variables used are declared and initialized as shown on the right. If a value cannot be determined, write "undefined" in the type column, and provide a brief explanation in the value column.

```
int a = 8, k = 9;
double b = 4.75;
bool flag1 = true, flag2 = false;
std::string phrase = "compute";
int arr[3] = {10, 2, 30};
float f = 5.55;
```

Expression	Type	Value
<code>static_cast<int>(b)+ a</code>	<code>int</code>	12
<code>arr[k-a] + 1</code>	<code>int</code>	3
<code>arr[arr[1]] + b</code>	<code>double</code>	34.75
<code>(1 + k / 3.0)* (f - 1.25)</code>	<code>double</code>	17.2
<code>(1.0 / k * f == f * 1.0 / k)</code>	<code>bool</code>	<code>true</code>
<code>!(flag1 flag2)&& (a == 8 b > 4)</code>	<code>bool</code>	<code>false</code>
<code>!((a <= k)&& (k <= a))</code>	<code>bool</code>	<code>true</code>
<code>phrase + std::to_string(2*a)</code>	<code>std::string</code>	"compute16"

Question 2: Tracing 28 marks

Find the output of the following programs.

(a) [2 marks]

```
1 #include<iostream>
2 int main() {
3     int x = 5;
4     while (x--) {
5         if (x % 2 == 0)
6             std::cout << "Even ";
7         else
8             std::cout << "Odd ";
9     }
10 }
```

Solution:
Even Odd
Even Odd
Even

(b) [2 marks]

```
1 #include<iostream>
2 int main() {
3     int arr[] = {1, 2, 3, 4};
4     for (int i = 0; i < 4; i++) {
5         std::cout << arr[i] << " ";
6         arr[i] += i;
7     }
8 }
```

Solution:
1 2 3 4

(c) [2 marks]

```
1 #include<iostream>
2 int main() {
3     int x = 5;
4     while (x > 0) {
5         if (x == 3) continue;
6         std::cout << x << " ";
7         x--;
8     }
9 }
```

Solution:
5 4
(stuck in infinite loop)

(d) [2 marks]

```
1 #include<iostream>
2 int main() {
3     int x = 5;
4     int y = 0;
5     while (x > 0) {
6         while (y < 3) {
7             if (y == 1) continue;
8             std::cout << x << " " << y << " ";
9             y++;
10        }
11        x--;
12    }
13 }
```

Solution:
5 0
(stuck in infinite loop)

(e) [2 marks]

```
1 #include<iostream>
2 int main() {
3     int x = 1;
4     switch (x) {
5         case 1:
6         case 2:
7             std::cout << "First ";
8         case 3:
9             std::cout << "Second ";
10        case 4:
11            std::cout << "Third ";
12            break;
13        default:
14            std::cout << "Default ";
15    }
16 }
```

Solution:
First Second
Third

(f) [3 marks]

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int x = 10, y = 5;
5     while (x > 5) {
6         int sum = 0;
7         for (int i = 1; i <= y; ++i) {
8             sum += i;
9         }
10        cout << sum << " ";
11        x -= 2;
12        y -= 1;
13    }
14 }
```

Solution:
15 10 6

(g) [3 marks]

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int arr[4] = {1, 2, 3, 4};
5     int x = 0, y = 1;
6     while (x < 4) {
7         if (arr[x] % 2 == 0)
8             y *= arr[x];
9         else
10            y += arr[x];
11        cout << y << " ";
12        x++;
13    }
14 }
```

Solution:
2 4 7 28

(h) [3 marks]

```

1 #include <iostream>
2 #include <string>
3 int main() {
4     std::string input_string = "Introduction to Programming";
5     std::string result = "";
6     // toupper(c)/tolower(c) convert char c to uppercase/lowercase
7     for (int i = 0; i < input_string.length(); i++) {
8         if (i % 2 == 0) {
9             result += toupper(input_string[i]);
10        } else {
11            result += tolower(input_string[i]);
12        }
13    }
14    std::cout << result << std::endl;
15    return 0;
16 }

```

Solution:

InTrOdUcTiOn tO

PrOgRaMmInG

(i) [3 marks]

```

1 #include<iostream>
2 using namespace std;
3 int main() {
4     int n = 5;
5     for (int i = 0; i < n; ++i) {
6         for (int j = 0; j < n; ++j) {
7             if ((i + j) % 2 == 0)
8                 cout << "#";
9             else
10                cout << "@";
11        }
12        cout << endl;
13    }
14 }

```

Solution:

#@#@#

@#@#@

#@#@#

@#@#@

#@#@#

(j) [3 marks]

```

1 #include<iostream>
2 #include<string>
3 using namespace std;
4 int main() {
5     string str = "pattern";
6     int count = 1;
7     for (int i = 0; i < str.length(); ++i) {
8         for (int j = 0; j < count; ++j) {
9             cout << str[i];
10        }
11        count += 2;
12        cout << endl;
13    }
14 }

```

Solution:

p

aaa

ttttt

ttttttt

eeeeeeeeee

rrrrrrrrrrrr

nnnnnnnnnnnnnnnnnn

(k) [3 marks]

```

1 #include<iostream>
2 using namespace std;
3 int main() {
4     bool a = false;
5     bool b = true;
6     bool c = false;
7     while (!a || b) {
8         if (b && !c) {
9             a = !a;
10        } else {
11            b = !b;
12        }
13        cout << a << " " << b << " " << c << " ";
14        c = !c;
15    }
16 }

```

Solution:

1 1 0 1 0 1

Question 3: Debugging 10 marks

- (a) [5 marks] Given sales data for two days, the following a program is supposed to output the total number of units sold. The sales data for each day is stored in an array. However, the program has a bug. Identify and fix the bug.

```

1 #include <iostream>
2 using namespace std;
3 int main() {
4     int day1_sales[] = {30, 25, 40, 35, 20, 45, 50, 55};
5     int day2_sales[] = {35, 40, 45, 50, 55, 60, 65};
6     int total_units_sold = 0;
7     for (int hour = 0; hour < size(day1_sales); hour++) {
8         total_units_sold += day1_sales[hour] + day2_sales[hour];
9     }
10    cout << total_units_sold << endl;
11    return 0;
12 }

```

Solution: The two arrays have different sizes. One solution is to use two loops. Another is to use the maximum size of the two arrays and check if the index is within the bounds of each array before accessing the element. Following is the code for the latter solution:

```

    int max_size = max(size(day1_sales), size(day2_sales));
    for (int hour = 0; hour < max_size; hour++) {
        if (hour < size1) total_units_sold += day1_sales[hour];
        if (hour < size2) total_units_sold += day2_sales[hour];
    }

```

- (b) [5 marks] You are given the following code that attempts to rotate an array of integers by moving each element one position to the right and wrapping the last element around to the first position. However, the code contains a bug, and it doesn't produce the expected result. Identify and fix the bug.

Example Input: 1 2 3 4 5

Expected Output: 5 1 2 3 4

```
1 #include <iostream>
2 int main() {
3     int arr[] = {1, 2, 3, 4, 5};
4     int size = sizeof(arr) / sizeof(arr[0]);
5
6     for (int i = 0; i < size; i++) {
7         // Move each element one position to the right
8         arr[i] = arr[(i + 1) % size];
9     }
10    // Output the array after "rotation"
11    cout << "Rotated array: ";
12    for (int i = 0; i < size; i++) {
13        std::cout << arr[i] << " ";
14    }
15    return 0;
16 }
```

Solution: The bug is in the assignment statement `arr[i] = arr[(i + 1) % size];` which actually moves each element one position to the left (and first element is overwritten). To fix the bug, we need to start from the last element and move each element one position to the right. We also need to store the last element before the loop starts. Following is the corrected code:

```
int last = arr[size - 1];
for (int i = size - 1; i > 0; i--) {
    arr[i] = arr[i - 1];
}
arr[0] = last;
```

Question 4: Coding20 marks(a) [10 marks] Grading System with Bonus Points:

Write a C++ program that calculates a student's final grade based on their exam score, participation in extracurricular activities, and behavior. The program should prompt the user to enter: (i) Their exam score [0-100], (ii) Whether they participated in extracurricular activities [Yes/No], and (iii) Whether they had good behavior during the term [Yes/No]. The final grade should be calculated as follows:

- If the exam score is less than 50, the final grade is 'F' regardless of other factors.
- If the exam score is 90 or above, the final grade is 'A'.

- If the exam score is between 50 and 89, the grade starts as follows:

50-59:	'D'
60-69:	'C'
70-79:	'B'
80-89:	'A-'

- Bonus Points:
 - Add 5 extra points to the exam score if the student participated in extracurricular activities.
 - Add 5 extra points if the student had good behavior.
 - The exam score after bonuses cannot exceed 100.

Display the final grade, including any bonus points applied. Following is a sample run:

```
Enter your exam score: 76
Did you participate in extracurricular activities? Yes
Did you have good behavior during the term? No

Final exam score (after bonus): 81
Your final grade is A-.
```

Solution:

```
#include <iostream>
#include <string>
using std::cin, std::cout, std::string, std::endl;

int main() {
    int exam_score;
    string extracurricular, behavior;
    cout << "Enter your exam score: ";
    cin >> exam_score;
    cout << "Did you participate in extracurricular activities? ";
    cin >> extracurricular;
    cout << "Did you have good behavior during the term? ";
    cin >> behavior;

    if (exam_score < 50) {
        cout << "Your final grade is F." << endl;
    } else {
        int bonus = 0;
        if (extracurricular == "Yes") bonus += 5;
        if (behavior == "Yes") bonus += 5;
```

```
    exam_score += bonus;
    if (exam_score > 100) exam_score = 100;
    char grade;
    if (exam_score >= 90) {
        grade = 'A';
    } else if (exam_score >= 80) {
        grade = 'A';
    } else if (exam_score >= 70) {
        grade = 'B';
    } else if (exam_score >= 60) {
        grade = 'C';
    } else {
        grade = 'D';
    }
    cout << "Final exam score (after bonus): " << exam_score << endl;
    cout << "Your final grade is " << grade << "." << endl;
}
return 0;
}
```


- (b) [10 marks] Given a 2D array (matrix) of integers, write a C++ program that prints the array in a snake-like pattern. The snake-like pattern means that the printing starts from the first row, going left to right, then the second row goes right to left, then the third row left to right, and so on. Your program should work for any array `arr` of size $m \times n$. For the following array, output should be:

1 2 3 4 8 7 6 5 9 10 11 12 16 15 14 13

```
#include <iostream>
using namespace std;
int main() {
    int arr[4][4] = { { 1, 2, 3, 4},
                      { 5, 6, 7, 8},
                      { 9, 10, 11, 12},
                      {13, 14, 15, 16} };

    // Your code here
```

Solution:

```
int m = size(arr), n = size(arr[0]);
for (int i = 0; i < m; i++) {
    if (i % 2 == 0) {
        for (int j = 0; j < n; j++) {
            cout << arr[i][j] << " ";
        }
    } else {
        for (int j = n - 1; j >= 0; j--) {
            cout << arr[i][j] << " ";
        }
    }
    cout << endl;
```

```
return 0;
}
```